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Prof. Dr. Peter Robin Hiesinger
Freie Universität Berlin
Fachbereich Biologie, Chemie, Pharmazie — Institut für Biologie
Königin-Luise-Str. 1–3
14195 Berlin (Dahlem)

Re: Wiss. Mitarbeiter*in, AG Neurobiologie — Kennung 21259000_2026

Dear Professor Hiesinger,

I am writing regarding the research-associate position in your group. The position explicitly includes the group's complementary computational work — digital image analysis, statistical evaluation of imaging data, and transcriptome and connectome analysis — and that is precisely where my work sits. My background is computational, and it is on that side that I would contribute to your group's imaging and analysis of Drosophila brains.

Digital image analysis of microscopy data. This is my strongest point of overlap. I have built a quantitative image-analysis framework for fluorescence and phase-contrast microscopy — segmentation, point-spread-function deconvolution, scale-field and coordinate reconstruction, and morphometry with uncertainty quantification, GPU-accelerated and available as a live tool (hieronymus-omega.vercel.app). For a group whose technical focus is live imaging of Drosophila brains, rigorous, reproducible image quantification is something I could contribute immediately.

Connectome and computational neuroscience. My independent research includes neural-circuit modelling: a closed-circuit decomposition of motor and neural signals into supraspinal and peripheral contributions, applied to neurodegeneration (Parkinson's, ataxia), together with circuit-graph (connectome-style) network analysis. This is directly relevant to the group's connectome and computational-modelling work.

Transcriptome analysis. I have developed pipelines integrating transcriptomics with genotype and metagenomics, validated against 1000 Genomes, gnomAD, and UK Biobank.

How I would contribute. My strength is the quantitative and computational side: turning imaging and sequencing data into rigorous, reproducible, statistically sound results, and building the tools to do so. I work closely and naturally with experimental groups, complementing their imaging and genetics rather than duplicating it. English is native; I have been resident in Germany since 2011.

If the computational and image-analysis side of the position is one you would consider filling with a computational specialist, I would welcome a conversation.

Yours sincerely,

Kundai Farai Sachikonye